SUBCRITICAL LIOUVILLE THEOREMS FOR WEIGHTED POLYHARMONIC LANE–EMDEN SYSTEMS

ABSTRACT. Let m, N be positive integers with $N > 2m$, let $p, q \geq 1$, and assume $pq > 1$. We prove that the weighted polyharmonic Lane–Emden system $(-\Delta)^m u = |x|^A v^p$, $(-\Delta)^m v = |x|^B u^q$ has no nonzero nonnegative solution continuous on $\mathbb{R}^N$ and classical away from the origin whenever $(N + A)/(p + 1) + (N + B)/(q + 1) > N - 2m$. The weights A, B are arbitrary real numbers. No boundedness, decay, radial symmetry, finite energy, or sign conditions on intermediate Laplacians are assumed. We first obtain an exact Riesz representation and show that a hypothetical nonzero solution forces $A, B > -2m$. We then adapt the localized source-layer pressure method for the second-order system to homogeneous Riesz kernels. A weighted virial identity and interval comparison give a universal local energy bound; subcritical scaling contradicts this bound. The unweighted higher-order Lane–Emden theorem follows as a corollary.

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1. INTRODUCTION

We consider the equal-order weighted system

$$(-\Delta)^m u = |x|^A v^p, \qquad (-\Delta)^m v = |x|^B u^q \quad \text{in } \mathbb{R}^N \setminus \{0\}, \tag{1.1}$$

where m, N are positive integers with $N > 2m$. The relevant subcritical inequality is

$$\frac{N + A}{p + 1} + \frac{N + B}{q + 1} > N - 2m. \tag{1.2}$$

For singular weights, the equations are imposed away from the origin, while continuity at the origin is part of the solution class. Our main result treats arbitrary real weights in this class.

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Theorem 1.1. *Let m, N be positive integers with $N > 2m$, let $p, q \geq 1$ satisfy $pq > 1$, and let $A, B \in \mathbb{R}$. Assume (1.2). If*

$$u, v \in C(\mathbb{R}^N) \cap C^{2m}(\mathbb{R}^N \setminus \{0\})$$

are nonnegative and satisfy (1.1) pointwise for every $x \neq 0$, then $u \equiv v \equiv 0$.

In particular, the theorem includes globally C^{2m} solutions. It requires no bound at infinity and no assumptions on the signs of $(-\Delta)^i u$ or $(-\Delta)^i v$, $1 \leq i < m$. The restrictions $A, B > -2m$ arise inside the proof as necessary conditions for a nonzero pair; they are not hypotheses of the theorem.

Corollary 1.2 (Unweighted higher-order Lane–Emden theorem)**.** *Let m, N be positive integers with $N > 2m$, let $p, q \geq 1$, and assume $pq > 1$. If*

$$\frac{1}{p+1} + \frac{1}{q+1} > \frac{N-2m}{N},$$

then every nonnegative classical pair $u, v \in C^{2m}(\mathbb{R}^N)$ satisfying

$$(-\Delta)^m u = v^p, \qquad (-\Delta)^m v = u^q \quad \text{in } \mathbb{R}^N$$

is identically zero.

Proof. Apply Theorem 1.1 with $A = B = 0$. □

Remark 1.3. Continuity at zero is essential for the stated weighted assertion. For example, when $N = 5$, $m = 2$, $p = q = 2$, and $A = B = -13/4$, the pair

$$u(x) = v(x) = \frac{297}{256}|x|^{-3/4}$$

is positive and smooth off zero and satisfies (1.1). Indeed, $\Delta^2|x|^t = t(t+3)(t-2)(t+1)|x|^{t-4}$ in dimension five. The weighted sum in (1.2) is $7/6 > 1 = N - 2m$, but this pair is singular at zero.

1.1. **Background and relation to earlier results.** The Lane–Emden problem asks how the critical hyperbola controls the existence of positive entire solutions of a coupled elliptic system. In the second-order unweighted case, Souplet proved the conjecture in four dimensions and obtained further nonexistence regions in higher dimensions [7]. Weighted second-order systems have also been studied through dimension-dependent and conditional results. Phan obtained Liouville theorems with boundedness or additional parameter hypotheses [6]; Li and Zhang removed the boundedness assumption in the three-dimensional weighted problem [8]. Li established further results for restricted exponent ranges in dimensions four and five [9]. More recently, Huang and Zou formulated the arbitrary-real-weight problem for solutions continuous at the origin and proved results under stability, prescribed decay, or rescaled annular energy conditions, with the additional parameter hypotheses specified in their theorems [10].

For higher order, an important direct predecessor is the work of F. Arthur, X. Yan and M. Zhao in *Discrete and Continuous Dynamical Systems* [2]. Their Theorem 1.1 proves unweighted nonexistence for positive entire C^{2m} solutions, with $p, q \geq 1$, $pq > 1$, under the strict hyperbola and the additional condition

$$\max\left\{\frac{2m(p+1)}{pq-1}, \frac{2m(q+1)}{pq-1}\right\} > N - 2m - 1. \tag{1.3}$$

They consequently obtain the full subcritical range in dimensions $N = 2m+1$ and $N = 2m+2$. Their theorem does not impose boundedness or decay. Corollary 1.2 removes (1.3) from the equal-order problem.

For nonnegative weights, Fazly proved subcritical nonexistence for radial solutions in all dimensions and for bounded nonnegative solutions when $N = 2m+1$ [4]. Phan obtained further polyharmonic results with boundedness and additional exponent conditions, as well as results without boundedness under a stronger unweighted subcritical hypothesis and an additional exponent condition [5]. Arthur and Yan proved weighted higher-order Liouville theorems under

prescribed slow decay and an additional scaling-exponent condition; they also obtained results without prescribed decay under stronger exponent assumptions [3]. These works show the roles played by radial symmetry, dimension, growth information and exponent restrictions in earlier approaches. Theorem 1.1 treats the strict weighted hyperbola in every dimension $N > 2m$, for the powers and solution class stated above.

The recent preprint *The Subcritical Hénon–Lane–Emden Conjecture*, dated September 24, 2026, presents a proof of the full second-order subcritical assertion for all positive powers and arbitrary real weights [11]. Its localized potential virial identity, interval-pair comparison and source-layer pressure argument are the methodological starting point of this paper. We adapt that pressure method to the Riesz kernel of order $2m$, and include the complete adapted argument below. Our higher-order theorem retains $p, q \geq 1$ and $pq > 1$.

1.2. **Proof mechanism and organization.** Two issues must be resolved before the potential argument applies. First, arbitrary real weights need not give locally integrable sources a priori. Continuity at the origin and $N > 2m$ allow a cutoff argument to recover local integrability and the distributional equations without an origin defect. Second, an entire polyharmonic equation can leave a polynomial remainder after inversion. Source bounds at every scale, combined with the strict hyperbola, give positive scaling exponents and eliminate that remainder. The resulting positive, finite potentials force $A, B > -2m$.

For the exact potential system, homogeneity supplies universal source mass bounds. A localized weighted virial identity relates the two local energies to a kernel interaction. Its cutoff loss is controlled by slicing source level sets into intervals and comparing the potentials with their endpoint averages. Narrow angular caps align the interval direction with the radial virial direction, while short-component retention and a polar-coordinate cone estimate control discarded layers. The weighted endpoint values are transported by the factors $|x|^{-B/q}$ and $|x|^{-A/p}$. The interval estimate depends only on the positive homogeneity exponent $N - 2m$, which makes the higher-order adaptation possible. The pressure estimate yields a universal local weighted energy bound, and strict subcritical scaling then gives the contradiction.

The proof is arranged in this dependency order. Section 2 removes the origin and establishes the exact Riesz representation. Section 3 proves universal estimates in a potential setting allowing any $1 < \sigma < N$. Section 4 derives the weighted virial identity and its localization errors. Section 5 proves the interval comparison and the source-layer pressure bound. Section 6 obtains the energy bound, applies scaling, and completes the proof of Theorem 1.1.

Throughout, $B_R = \{x \in \mathbb{R}^N : |x| < R\}$ and $t_+ = \max\{t, 0\}$. All integrals are Lebesgue integrals. An integral without a displayed domain is over $\mathbb{R}^N$; an interaction integral is over $\mathbb{R}^N \times \mathbb{R}^N$.

2. From the differential system to exact potentials

Throughout this section, let $\sigma = 2m < N$, $\lambda = N - \sigma$, and

$$\alpha = \frac{\sigma + A + p(\sigma + B)}{pq - 1}, \qquad \beta = \frac{\sigma + B + q(\sigma + A)}{pq - 1}. \tag{2.1}$$

The identities

$$p\beta = \alpha + \sigma + A, \qquad q\alpha = \beta + \sigma + B \tag{2.2}$$

will be used repeatedly. We write

$$g(x) = |x|^A v(x)^p, \qquad f(x) = |x|^B u(x)^q \quad (x \neq 0),$$

and give both sources the value zero at the origin. Values on this singleton do not affect any integral.

2.1. Removal of the origin.

Lemma 2.1. *Suppose $w \in C(\mathbb{R}^N) \cap C^{2m}(\mathbb{R}^N \setminus \{0\})$, and $G \geq 0$ is continuous off the origin, with $(-\Delta)^m w = G$ there. Then $G \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$, and the equation holds in distributions on $\mathbb{R}^N$.*

Proof. Choose $0 \le \chi \le 1$ in $C_c^\infty(B_2)$, equal to one on B_1, and put $\chi_\varepsilon(x) = \chi(x/\varepsilon)$. For $\zeta \in C_c^\infty(\mathbb{R}^N)$, testing away from zero gives

$$\int G\zeta(1-\chi_\varepsilon) = \int w(-\Delta)^m\zeta - \int w(-\Delta)^m(\zeta\chi_\varepsilon).$$

For $0 < \varepsilon \le 1$, a derivative of order i falling on χ_ε contributes a factor ε^{-i}, and its support has volume $O(\varepsilon^N)$. The product rule therefore bounds the last integral by

$$C_\zeta \sup_{B_2} |w| \sum_{i=0}^{\sigma} \varepsilon^{N-i} \le C_{\zeta,w}\varepsilon^{N-\sigma} \longrightarrow 0.$$

Given $R > 0$, first choose $\zeta \ge 0$ equal to one on B_R. The nonnegative integrands $G\zeta(1-\chi_\varepsilon)$ converge almost everywhere to $G\zeta$. Fatou's lemma and the test identity give

$$\int_{B_R} G \le \int G\zeta \le \liminf_{\varepsilon\downarrow 0} \int G\zeta(1-\chi_\varepsilon) = \int w(-\Delta)^m\zeta < \infty.$$

Thus $G \in L^1_{\mathrm{loc}}$. For an arbitrary signed test function ζ, the bound $|G\zeta(1-\chi_\varepsilon)| \le G|\zeta| \in L^1$ now permits dominated convergence in the same identity, giving $\int G\zeta = \int w(-\Delta)^m\zeta$. □

2.2. **Source bounds at every scale.**

Lemma 2.2. *Let $p, q \ge 1$, $pq > 1$, and $A, B \in \mathbb{R}$. A nonnegative pair satisfying the differential system and the regularity in Theorem 1.1 obeys, for every $R > 0$,*

$$\int_{B_R} g \le CR^{\lambda-\alpha}, \qquad \int_{B_R} f \le CR^{\lambda-\beta}, \tag{2.3}$$

$$\int_{B_{2R}\setminus B_R} u \le CR^{N-\alpha}, \qquad \int_{B_{2R}\setminus B_R} v \le CR^{N-\beta}. \tag{2.4}$$

Here and below a universal constant is independent of the solution. The constants in this lemma depend only on N, m, p, q, A, B.

Proof. Lemma 2.1 makes both sources locally integrable and both equations valid in distributions. Put $D_R = B_{2R} \setminus B_R$. We shall use the following annular estimate, valid for $k \ge 1$, $\mu \in \mathbb{R}$, and a nonnegative measurable H:

$$\int_{D_R} H \le CR^{N(1-1/k)-\mu/k} \left(\int_{D_R} |x|^\mu H^k\right)^{1/k}. \tag{2.5}$$

For $k > 1$, this is Hölder's inequality, since

$$\left(\int_{D_R} |x|^{-\mu/(k-1)}\,dx\right)^{1-1/k} = C_{N,k,\mu}R^{N(1-1/k)-\mu/k}.$$

The constant is finite for every real μ, because the rescaled annulus stays away from zero. For $k = 1$, use instead $\sup_{D_R} |x|^{-\mu} \le C_\mu R^{-\mu}$.

Fix $0 \le \varphi \le 1$ in $C_c^\infty(B_2)$, equal to one on B_1, choosing this cutoff once depending only on N, and put $\varphi_R(x) = \varphi(x/R)$. Choose integers $s, t \ge \sigma$, depending only on p, q, σ, such that

$$q(t-\sigma) \ge s, \qquad p(s-\sigma) \ge t.$$

Indeed, for sufficiently large integer s, the interval $[\sigma + s/q, p(s-\sigma)]$ has length $(p-1/q)s - (p+1)\sigma \ge 1$, since $pq > 1$. Take an integer t in that interval. Every term of the product rule differentiates at most σ factors of φ_R, leaving at least $t-\sigma$ undifferentiated factors. Since $0 \le \varphi_R \le 1$, this gives

$$|(-\Delta)^m\varphi_R^t| \le CR^{-\sigma}\varphi_R^{t-\sigma}\mathbf{1}_{D_R},$$

and the same estimate holds with s. The quantities $F_R = \int g\varphi_R^t$ and $G_R = \int f\varphi_R^s$ are finite by local source integrability. For $H = u\varphi_R^{t-\sigma}$, the weighted moment in (2.5), with $k = q$ and

$\mu = B$, satisfies $\int_{D_R} f\varphi_R^{q(t-\sigma)} \le G_R$, because $q(t-\sigma) \ge s$ and $0 \le \varphi_R \le 1$. The component swap gives $\int_{D_R} g\varphi_R^{p(s-\sigma)} \le F_R$. Testing the two equations and applying these bounds yields

$$F_R \le CR^{-\sigma} \int_{D_R} u\varphi_R^{t-\sigma} \le CR^{-\sigma+N(1-1/q)-B/q} G_R^{1/q},$$
$$G_R \le CR^{-\sigma+N(1-1/p)-A/p} F_R^{1/p}.$$

The endpoint $k = 1$ in (2.5) covers $q = 1$ or $p = 1$ in these same inequalities.

If either F_R or G_R is zero, both are zero. Otherwise substitution and division yield

$$F_R^{1-1/(pq)} \le CR^{N(1-1/(pq))-\sigma(1+1/q)-B/q-A/(pq)}.$$

Raising this inequality to $pq/(pq-1)$ gives

$$F_R \le CR^{N-[\sigma p(q+1)+A+pB]/(pq-1)} = CR^{\lambda-\alpha}.$$

The corresponding calculation gives $G_R \le CR^{\lambda-\beta}$. Since the cutoff equals one on B_R, this proves (2.3). Finally, apply (2.5) with $H = u$, $k = q$, and $\mu = B$, and use the source bound at radius $2R$:

$$\int_{D_R} u \le CR^{N(1-1/q)-B/q} \left(\int_{B_{2R}} f \right)^{1/q} \le CR^{N-(\sigma+\beta+B)/q} = CR^{N-\alpha},$$

where (2.2) was used in the last equality. Interchanging the components gives the estimate for v. All cutoffs and integers were fixed independently of R, u, v; the constants in (2.3) and (2.4) therefore have the asserted dependence. □

2.3. **The polynomial remainder vanishes.** Put

$$K_\sigma(z) = c_{N,\sigma}|z|^{\sigma-N}, \qquad c_{N,\sigma} = \frac{\Gamma((N-\sigma)/2)}{2^\sigma \pi^{N/2}\Gamma(\sigma/2)}. \tag{2.6}$$

Proposition 2.3. *Under the hypotheses of Theorem 1.1, a nonzero pair is strictly positive everywhere, necessarily satisfies $A, B > -\sigma$, and has the finite representations*

$$u = K_\sigma * g, \qquad v = K_\sigma * f \quad \text{at every point of } \mathbb{R}^N. \tag{2.7}$$

Proof. If either component is identically zero, the equations off the origin and continuity imply that both are zero. Otherwise each component is positive on some ball disjoint from zero, so each source has positive mass in a sufficiently large fixed ball. Sending R to infinity in (2.3) shows that $\alpha, \beta \le \lambda$. For

$$a = \frac{1}{q+1}, \qquad b = \frac{1}{p+1}, \qquad \gamma = a(N+B) + b(N+A) - \lambda,$$

direct algebra gives

$$\gamma = (1-a-b)(\alpha+\beta+\sigma-N), \qquad 1-a-b = \frac{pq-1}{(p+1)(q+1)} > 0. \tag{2.8}$$

The strict inequality in the main theorem is $\gamma > 0$. Consequently $\alpha + \beta > \lambda$, and hence $0 < \alpha, \beta \le \lambda$. In particular $N-\alpha, N-\beta \ge \sigma > 0$. For example, summing (2.4) over the disjoint annuli $B_{2^{-i}R} \setminus B_{2^{-i-1}R}$, $i \ge 0$, gives

$$\int_{B_R} u \le CR^{N-\alpha} \sum_{i\ge 0} 2^{-(i+1)(N-\alpha)} \le CR^{N-\alpha},$$

because $N - \alpha \ge \sigma > 0$; the singleton at zero has measure zero. The same calculation for v gives

$$\int_{B_R} u \le CR^{N-\alpha}, \qquad \int_{B_R} v \le CR^{N-\beta} \quad (R > 0). \tag{2.9}$$

The availability of the annular estimate at small as well as large scales makes these constants universal.

Define $U = K_\sigma * g$, initially allowing an infinite value at zero. By dyadic summation of the source bound,

$$\begin{aligned}\int_{|y|>R} |y|^{-\lambda} g(y)\,dy &\le \sum_{i\ge 0} (2^i R)^{-\lambda} \int_{B_{2^{i+1}R}} g \\ &\le CR^{-\alpha} \sum_{i\ge 0} 2^{-i\alpha} \le CR^{-\alpha}.\end{aligned} \tag{2.10}$$

The corresponding tail bound for f has β in place of α. To justify finiteness and continuity away from zero, fix $x_0 \neq 0$ and choose $0 < \rho < |x_0|/4$. The source g is bounded on a fixed neighborhood of x_0, and for $|x - x_0| \le \rho$ the local part satisfies

$$\int_{B_{2\rho}(x_0)} K_\sigma(x-y) g(y)\,dy \le C_{x_0,u,v} \int_{B_{3\rho}} |z|^{-\lambda}\,dz \le C_{x_0,u,v}\rho^\sigma.$$

The constant can be fixed for all sufficiently small ρ. On the complementary region, $|x-y| \ge \rho$; local source integrability and (2.10) provide an integrable dominating function for x near x_0. This part is finite and continuous by dominated convergence. For fixed ρ, the local contributions at x and x_0 are bounded by the displayed estimate, so first letting $x \to x_0$ and then $\rho \downarrow 0$ proves continuity of U.

For $y \in B_{2R}$, the inclusion $B_R - y \subset B_{3R}$ gives $\int_{B_R} K_\sigma(x-y)\,dx \le CR^\sigma$. For $|y| > 2R$ and $x \in B_R$, use $|x-y| \ge |y|/2$. Tonelli's theorem therefore gives

$$\int_{B_R} U \le CR^\sigma \int_{B_{2R}} g + CR^N \int_{|y|>2R} |y|^{-\lambda} g(y)\,dy \le CR^{N-\alpha}.$$

Thus U is locally integrable.

We use the standard Newtonian fundamental solution: $c_{N,2} = 1/((N-2)|\mathbb{S}^{N-1}|)$ and $-\Delta K_2 = \delta_0$, with the sign convention in [1, Proposition 9.6]. Radial differentiation and the gamma recursion give $-\Delta K_{2i} = K_{2i-2}$ off zero for $2 \le i \le m$. Both sides are homogeneous distributions of degree $2i-2-N > -N$. The uniqueness of homogeneous extension [1, Theorem 5.6] makes this identity distributional on all of $\mathbb{R}^N$; iteration gives $(-\Delta)^m K_\sigma = \delta_0$.

Let $\zeta \in C_c^\infty(B_S)$, with $S \ge 1$, and put

$$J_\zeta(y) = \int K_\sigma(x-y)|(-\Delta)^m \zeta(x)|\,dx.$$

Local integrability of the kernel gives $J_\zeta(y) \le C_\zeta$ for $|y| \le 2S$, and $|x-y| \ge |y|/2$ gives $J_\zeta(y) \le C_\zeta |y|^{-\lambda}$ for $|y| > 2S$. Thus local source integrability and (2.10) imply $\int g J_\zeta < \infty$. Absolute Fubini and the translated fundamental-solution identity now yield

$$\begin{aligned}\int U(x)(-\Delta)^m\zeta(x)\,dx &= \int g(y) \left(\int K_\sigma(x-y)(-\Delta)^m\zeta(x)\,dx \right) dy \\ &= \int g(y)\zeta(y)\,dy.\end{aligned}$$

Hence $(-\Delta)^m U = g$ in distributions.

To see explicitly that u and U are tempered, choose an integer $M > N - \alpha$. Their nonnegative ball bounds imply

$$\int_{\mathbb{R}^N} (u+U)(1+|x|)^{-M}\,dx \le C + C\sum_{i\ge 0} 2^{i(N-\alpha-M)} < \infty.$$

Their action on a Schwartz function ζ is therefore bounded by a constant times $\sup_x (1+|x|)^M |\zeta(x)|$, a Schwartz seminorm. Hence $h = u - U$ is tempered and satisfies

$$(-\Delta)^m h = 0, \qquad \int_{B_R} |h| \le CR^{N-\alpha}.$$

Put $T = \widehat{h}$. If ζ has compact support disjoint from zero, then $\zeta/|\xi|^{2m}$ is a valid test function, so $T(\zeta) = 0$. Thus T is supported at zero and is a compactly supported distribution. By the structure theorem for such distributions [1, Theorem 4.19], T is a finite linear combination of derivatives of δ_0. Its inverse Fourier transform is a polynomial P, agreeing almost everywhere

with h. If P is nonzero, let $d \geq 0$ be its degree and P_d its leading homogeneous part. The change of variables $x = Rz$ and uniform convergence of $R^{-d}P(Rz)$ to $P_d(z)$ on B_1 give

$$R^{-N-d}\int_{B_R}|h| \longrightarrow \int_{B_1}|P_d| > 0,$$

whereas the ball bound makes the left side at most $CR^{-d-\alpha} \to 0$. Therefore $h = 0$. We obtain $u = U$ almost everywhere and, by continuity, at every nonzero point. Applying the same argument to $V = K_\sigma * f$ gives $v = V$ away from zero.

The positive kernel and nonzero sources imply positivity off zero. If $x_i \to 0$, $x_i \neq 0$, Fatou's lemma gives

$$0 < U(0) \leq \liminf_i U(x_i) = u(0) < \infty, \qquad 0 < V(0) \leq v(0) < \infty.$$

In particular $u(0), v(0) > 0$, before equality of the potentials at zero has been established. If $A \leq -\sigma$, continuity of v would give

$$U(0) \geq Cv(0)^p \int_0^\delta r^{\sigma+A-1}\,dr = \infty,$$

a contradiction. Thus $A > -\sigma$, and symmetrically $B > -\sigma$.

It remains to prove continuity of the potential at zero. At $x = 0$, boundedness of v near zero gives directly

$$\int_{|y|<\delta}|y|^{-\lambda+A}v(y)^p\,dy \leq C_{u,v}\int_0^\delta r^{\sigma+A-1}\,dr = \frac{C_{u,v}}{\sigma+A}\delta^{\sigma+A}.$$

For $0 < |x| \leq \delta/2$, split the same local potential into $|y| < 2|x|$ and $2|x| \leq |y| < \delta$. In the first region, writing $r = |x|$ and $e = x/r$ gives

$$\int_{|y|<2r}|x-y|^{-\lambda}|y|^A v(y)^p\,dy \leq C_{u,v}r^{\sigma+A}\int_{|z|<2}|e-z|^{-\lambda}|z|^A\,dz \leq C_{u,v}r^{\sigma+A}.$$

The last integral is uniformly bounded for $|e| = 1$: near $z = 0$ the other factor is bounded and $A > -\sigma > -N$; near $z = e$ the weight is bounded and $\lambda < N$; on the remaining compact region both factors are bounded. Rotational invariance gives uniformity in e. In the second region, $|x - y| \geq |y|/2$, so radial integration gives a bound $C_{u,v}\delta^{\sigma+A}$. Together, these estimates yield

$$\sup_{|x|\leq\delta/2}\int_{|y|<\delta}|x-y|^{-\lambda}|y|^A v(y)^p\,dy \leq C_{u,v}\delta^{\sigma+A} \longrightarrow 0.$$

The part outside B_δ is continuous at zero by dominated convergence and the tail estimate. Denote this outer part by $U_{\mathrm{out},\delta}$. For $|x| \leq \delta/2$, the local estimate at x and at zero gives

$$|U(x) - U(0)| \leq 2C_{u,v}\delta^{\sigma+A} + |U_{\mathrm{out},\delta}(x) - U_{\mathrm{out},\delta}(0)|.$$

First let $x \to 0$ with δ fixed, then let $\delta \downarrow 0$. This proves continuity of U at zero and hence $U(0) = u(0)$. The same argument proves $V(0) = v(0)$, completing (2.7). □

3. Uniform estimates for the potential system

For the rest of the proof we work in a more general potential setting. Let $N \geq 2$ be an integer, let $1 < \sigma < N$, $c > 0$, $A, B > -\sigma$, and let $p, q \geq 1$ satisfy $pq > 1$. Set $\lambda = N - \sigma$ and $K(z) = c|z|^{-\lambda}$. Assume that $u, v : \mathbb{R}^N \to (0, \infty)$ are finite and continuous and satisfy

$$u(x) = \int K(x-y)g(y)\,dy, \qquad v(y) = \int K(x-y)f(x)\,dx, \quad f = |x|^B u^q, \quad g = |x|^A v^p. \tag{3.1}$$

As in Section 2, set $f(0) = g(0) = 0$; these singleton values have no effect on the potential equations. The exponents α, β are still defined by (2.1); now both are positive because $\sigma + A, \sigma + B > 0$ and $pq - 1 > 0$. The algebraic identities (2.2) hold for this general σ as well. All constants called universal in this setting depend only on N, σ, p, q, A, B, c, and on any auxiliary parameter explicitly indicated.

Lemma 3.1. *Every pair in (3.1) satisfies*

$$\int_{B_R} g \le CR^{\lambda-\alpha}, \qquad \int_{B_R} f \le CR^{\lambda-\beta} \quad (R>0), \tag{3.2}$$

and

$$\int_{B_4} (f+g+u+v) \le C. \tag{3.3}$$

For $x \in B_2$ and $0<\ell\le 1$,

$$\int_{|x-y|\ge\ell} K(x-y)(f(y)+g(y))\,dy \le C\ell^{-\lambda}. \tag{3.4}$$

Moreover $u,v \in C^1(\mathbb{R}^N\setminus\{0\})$, and for every $x\neq 0$ the following integrals converge absolutely:

$$\nabla u(x) = \int \nabla K(x-y)g(y)\,dy, \qquad \nabla v(x) = \int \nabla K(x-y)f(y)\,dy. \tag{3.5}$$

The sources f,g are also C^1 off zero.

Proof. Let $M_f = \int_{B_1} f$, $M_g = \int_{B_1} g$. Continuity bounds u,v on $\overline{B}_1$, and $A,B > -\sigma > -N$ makes both weights integrable there. Thus M_f, M_g are finite; positivity of the pair makes both strictly positive. For $x,y\in B_1$, $K(x-y)\ge c2^{-\lambda}$. Writing $C_0 = c2^{-\lambda}$, we obtain

$$M_f \ge C_0^q M_g^q \int_{B_1} |x|^B\,dx, \qquad M_g \ge C_0^p M_f^p \int_{B_1} |x|^A\,dx.$$

The two coefficients

$$c_f = C_0^q \frac{|\mathbb{S}^{N-1}|}{N+B}, \qquad c_g = C_0^p \frac{|\mathbb{S}^{N-1}|}{N+A}$$

are positive and depend only on the fixed parameters. Substitution gives $M_f \ge c_f c_g^q M_f^{pq}$, and hence $M_f^{pq-1} \le (c_f c_g^q)^{-1}$. The first inequality also gives $M_g \le (M_f/c_f)^{1/q}$. Thus both masses have universal upper bounds, including when either exponent is one.

The dilations

$$u_R(x) = R^\alpha u(Rx), \qquad v_R(x) = R^\beta v(Rx) \tag{3.6}$$

preserve (3.1). Indeed, changing variables $z=Ry$ and using the homogeneity of K gives

$$\int K(x-y)|y|^A v_R(y)^p\,dy = R^{p\beta-\sigma-A}\int K(Rx-z)|z|^A v(z)^p\,dz = R^\alpha u(Rx) = u_R(x).$$

The other equation follows from $q\alpha-\sigma-B=\beta$ in exactly the same way. Their sources satisfy

$$f_R(x) = R^{\beta+\sigma} f(Rx), \qquad g_R(x) = R^{\alpha+\sigma} g(Rx).$$

Consequently

$$\int_{B_1} g_R = R^{\alpha-\lambda}\int_{B_R} g, \qquad \int_{B_1} f_R = R^{\beta-\lambda}\int_{B_R} f.$$

The dilated pair is again finite, positive and continuous. Applying the same universal unit mass bounds to it proves (3.2) for every $R>0$.

The dyadic calculation in (2.10) uses only the source mass bound and the positive exponents λ,α; it therefore applies to the present kernel and, by component exchange, to both sources:

$$\int_{|y|>R} |y|^{-\lambda} g(y)\,dy \le CR^{-\alpha}, \qquad \int_{|y|>R} |y|^{-\lambda} f(y)\,dy \le CR^{-\beta}. \tag{3.7}$$

The Tonelli estimate in the proof of Proposition 2.3 likewise uses only kernel integrability and these mass and tail bounds. Applying it at radius 4 gives

$$\int_{B_4} u \le C\int_{B_8} g + C\int_{|y|>8} |y|^{-\lambda} g(y)\,dy \le C,$$

and the same estimate holds for v. Together with the source bounds at radius 4, this proves (3.3).

For (3.4), fix $x \in B_2$, $0 < \ell \le 1$, and take $H = f$ or $H = g$. On B_8, the truncated kernel is at most $c\ell^{-\lambda}$; outside B_8, $|x-y| \ge |y|/2$. Hence

$$\int_{|x-y|\ge\ell} K(x-y)H(y)\,dy \le c\ell^{-\lambda}\int_{B_8} H + C\int_{|y|>8} |y|^{-\lambda}H(y)\,dy \le C\ell^{-\lambda}.$$

Here the source and tail bounds are universal, and $\ell^{-\lambda} \ge 1$ absorbs the constant tail contribution. Adding the two source estimates gives the claimed bound.

Finally

$$\nabla K(z) = -\lambda\,\frac{z}{|z|^2}K(z)$$

is locally integrable, since its radial integral near zero is a constant times $\int_0^1 r^{\sigma-2}\,dr < \infty$. The uniqueness of homogeneous extension used in Section 2 identifies this vector field with the distributional gradient: its components and the distributional derivatives of K agree off zero and have degree $\sigma - N - 1 > -N$.

We prove the regularity for either source $H \in \{f, g\}$, with $W = K * H$. Fix $x_0 \ne 0$, choose $0 < r < |x_0|/4$, and take $0 \le \chi \le 1$ in $C_c^\infty(B_{3r}(x_0))$, equal to one on $B_{2r}(x_0)$. Put $H_0 = \chi H$ and $H_1 = (1-\chi)H$. The source H_0 is bounded and compactly supported, and for $x \in B_r(x_0)$,

$$\int |\nabla K(x-y)|H_0(y)\,dy \le C\|H_0\|_\infty r^{\sigma-1} < \infty.$$

Compact-source convolution in distributions gives $\partial_j(K*H_0) = (\partial_j K)*H_0$. For $F_j = \mathbf{1}_{B_{5r}}\partial_j K \in L^1$, and for $x, x+h$ in $B_r(x_0)$, these derivative convolutions satisfy

$$|((\partial_j K) * H_0)(x+h) - ((\partial_j K) * H_0)(x)| \le \|H_0\|_\infty \|F_j(\,\cdot + h) - F_j\|_{L^1} \longrightarrow 0.$$

The same translation argument with the locally truncated K shows that $K * H_0$ is continuous. Thus this potential has continuous weak first derivatives. Local mollification and passage to the limit in the fundamental theorem of calculus identify them with its classical first derivatives, so $K * H_0$ is C^1 on $B_r(x_0)$.

On the support of H_1, $|x_0 - y| \ge 2r$, and for $x \in B_r(x_0)$ we have $|x-y| \ge |x_0-y|/2 \ge r$. Consequently

$$K(x-y) + |\nabla K(x-y)| \le C_r K(x_0-y), \qquad \int K(x_0-y)H_1(y)\,dy \le W(x_0) < \infty.$$

Dominated differentiation therefore makes $K*H_1$ a C^1 function on this ball, with an absolutely convergent gradient integral. Adding the two parts proves (3.5) and the claimed C^1 regularity, because $x_0 \ne 0$ was arbitrary. The local constants in this regularity argument may depend on the pair and on x_0; they are used only to justify differentiation. Finally, the weights are smooth off zero and u, v are positive there, so the chain rule gives $f, g \in C^1(\mathbb{R}^N \setminus \{0\})$. □

4. A localized weighted virial identity

We continue in the potential setting of Section 3. In particular, $K(z) = c|z|^{-\lambda}$, $\lambda = N - \sigma$, $f = |x|^B u^q$, $g = |x|^A v^p$, and $u = K * g$, $v = K * f$. Put

$$a = \frac{1}{q+1}, \qquad b = \frac{1}{p+1}, \qquad j_0 = N+3, \qquad k_0 = N(j_0+1), \qquad L_0 = k_0 + N + 2.$$

The constants in this section are independent of the particular potential pair. Bounds used solely to justify an integration may, however, depend on its local supremum norms.

Choose the radial cutoff

$$\begin{aligned} d(x) &= \min\{1, (2-|x|)_+\}, \quad & \eta(x) &= \big[(1-(|x|-1)_+^3)_+\big]^{L_0}, \\ w(x) &= -x\cdot\nabla\eta(x), & X(x) &= x\eta(x). \end{aligned} \tag{4.1}$$

Thus $\eta = 1$ on B_1, $\eta = 0$ outside B_2, and $0 \le \eta \le 1$. To verify its regularity and the needed bounds, write $r = |x|$, $s = r - 1$, and $Q = 1 - s^3$ on $1 < r < 2$. For the radial profile $H(r) = Q^{L_0}$,

$$H'(r) = -3L_0 s^2 Q^{L_0-1}, \qquad H''(r) = 9L_0(L_0-1)s^4 Q^{L_0-2} - 6L_0 s Q^{L_0-1}.$$

Both derivatives vanish at the inner junction $r = 1$; they also vanish at $r = 2$, since $L_0 > 2$. The profile is constant near zero, so $\eta, X \in C^2(\mathbb{R}^N)$. Moreover $w = -rH' \geq 0$, with support in $\{1 \leq |x| \leq 2\}$, and $w \in C^1(\mathbb{R}^N)$. The factorization $Q = d(x)(1 + s + s^2)$ gives $d \leq Q \leq 3d$. The radial Hessian formula

$$\nabla^2\eta = H''\omega_x \otimes \omega_x + \frac{H'}{r}(\mathrm{Id} - \omega_x \otimes \omega_x), \qquad \nabla w = -(H' + rH'')\omega_x$$

and the product rule for $X = x\eta$ therefore imply

$$d^{L_0} \leq \eta \leq Cd^{L_0}, \qquad |\nabla\eta| + w \leq Cd^{L_0-1}, \tag{4.2}$$

$$|\nabla^2\eta| + |\nabla w| + |\nabla^2 X| \leq Cd^{L_0-2}, \qquad dw \leq C\eta. \tag{4.3}$$

For the last estimate, use $dw = 3L_0 r s^2\eta/(1 + s + s^2) \leq 6L_0\eta$ on the annulus. Inside B_1 and outside B_2 the derivative bounds are immediate from the constant profiles, so the displayed estimates hold globally. Their constants depend only on N. The distance function d is globally 1-Lipschitz.

The nonnegative kernel measure

$$d\pi(x, y) = f(x)g(y)K(x - y)\,dx\,dy$$

satisfies the Tonelli identities

$$\int_{\mathbb{R}^N} f(x)g(y)K(x - y)\,dy = f(x)u(x), \qquad \int_{\mathbb{R}^N} f(x)g(y)K(x - y)\,dx = g(y)v(y). \tag{4.4}$$

We shall use

$$E_f = \int \eta f u, \qquad E_g = \int \eta g v, \qquad E = E_f + E_g, \qquad W_f = \int w f u, \qquad W_g = \int w g v.$$

They are finite because the potentials are locally bounded and $A, B > -\sigma > -N$. For $x \neq 0$, write $\omega_x = x/|x|$, and, for a unit vector e, set

$$D_e(x, y) = \frac{((x - y) \cdot e)^2}{|x - y|^2}, \qquad D(x, y) = D_{\omega_x}(x, y), \qquad \Phi(x, y) = \frac{(X(x) - X(y)) \cdot (x - y)}{|x - y|^2}. \tag{4.5}$$

Set the density of π and the geometric quantities to zero on the diagonal, and choose any unit vector ω_0. These changes are on Lebesgue null sets and do not affect the integrals. In particular, $0 \leq D_e, D \leq 1$. Since X is globally Lipschitz and vanishes outside B_2,

$$\iint |\Phi|\,d\pi \leq \|\nabla X\|_\infty \left(\int_{B_2} fu + \int_{B_2} gv\right) < \infty.$$

Here $\Phi = 0$ when both endpoints lie outside B_2, and the Tonelli identities give the two finite marginals in this bound.

Lemma 4.1 (Weighted virial identity)**.** *The following identity is absolutely convergent:*

$$a(N + B)E_f + b(N + A)E_g - aW_f - bW_g = \lambda \iint \Phi\,d\pi. \tag{4.6}$$

Proof. Use the gradient formulas (3.5), and put

$$H_g(x) = \int |\nabla K(x - y)|g(y)\,dy, \qquad H_f(x) = \int |\nabla K(x - y)|f(y)\,dy.$$

The local estimates in their proof bound H_f, H_g uniformly on each compact set away from zero, by a finite covering. We need also control their pairings with X at zero. For $0 < r = |x| < 1/2$ and $|x - y| \leq r/2$, the source satisfies $g(y) \leq C_{u,v}r^A$, since $r/2 \leq |y| \leq 3r/2$ and v is bounded on B_1. As $|X(x)| \leq r$,

$$|X(x)| \int_{|x-y|\leq r/2} |\nabla K(x - y)|g(y)\,dy \leq C_{u,v}r^{A+1} \int_0^{r/2} t^{\sigma-2}\,dt \leq C_{u,v}r^{A+\sigma}.$$

Here $\sigma > 1$ and $A + \sigma > 0$. On the complementary region, $r \leq 2|x - y|$, so

$$|X(x)| \int_{|x-y|>r/2} |\nabla K(x - y)|g(y)\,dy \leq 2\lambda u(x).$$

Both expressions are bounded near zero. Multiplication by the locally integrable source f proves integrability of $f|X|H_g$ there. Interchanging the components gives integrability of $g|X|H_f$, using $\sigma + B > 0$. Away from zero the preceding compact-set bounds apply, and X is supported in $\overline{B}_2$. Thus

$$\iint (|X(x)| + |X(y)|)|\nabla K(x-y)|f(x)g(y)\,dx\,dy < \infty.$$

This is the absolute Fubini bound for both gradient pairings below; the local constants used to establish it may depend on the pair.

For $0 < \rho < 1$, apply the divergence theorem on $B_3 \setminus \overline{B}_\rho$ to the C^1 vector field $|x|^B X u^{q+1}$. Its outer flux vanishes because $X = 0$ on ∂B_3. Since

$$\operatorname{div}(|x|^B X) = |x|^B((N+B)\eta - w),$$

the chain rule gives

$$\int_{|x|>\rho} fX\cdot\nabla u = -a\int_{|x|>\rho}((N+B)\eta - w)fu - a\rho^{B+1}\int_{\partial B_\rho} u^{q+1}\,dS.$$

The inner outward normal is $-x/\rho$, and $X = x$ on ∂B_ρ, giving the negative boundary term. Its absolute value is at most

$$a|\mathbb{S}^{N-1}|\sup_{B_1} u^{q+1}\rho^{N+B} \longrightarrow 0,$$

because $B > -N$. The absolute gradient pairing and the finite local energy integrals justify dominated convergence in the bulk terms. Letting $\rho \downarrow 0$ therefore gives

$$\int fX\cdot\nabla u = -a(N+B)E_f + aW_f.$$

Interchanging the components gives $\int gX\cdot\nabla v = -b(N+A)E_g + bW_g$, with inner boundary error $O_{u,v}(\rho^{N+A})$.

The kernel is even and its gradient is odd. In particular, $\nabla v(y) = -\int \nabla K(x-y)f(x)\,dx$. Absolute Fubini now gives

$$\begin{aligned}\int fX\cdot\nabla u + \int gX\cdot\nabla v &= \iint f(x)g(y)(X(x)-X(y))\cdot\nabla K(x-y)\,dx\,dy\\ &= -\lambda\iint \Phi\,d\pi.\end{aligned}$$

Combining this with the two integration-by-parts identities proves (4.6). □

Lemma 4.2 (Localization of the interaction)**.** *For every $\varepsilon > 0$ there is a universal constant C_ε such that*

$$|E_f - E_g| + \iint |\Phi - \eta(x) + w(x)D(x,y)|\,d\pi \le \varepsilon E + C_\varepsilon. \tag{4.7}$$

Proof. Work off the π-null diagonal. Fix $0 < \delta < 1/4$, and assign to each point with $d(z) > 0$ the length $\ell(z) = \delta d(z)^{j_0}$. Call a pair (x,y) close if $|x-y| \le \ell(z)$ for at least one endpoint $z \in \{x,y\}$ with $d(z) > 0$; all other pairs are far. The close set is the union of the two Borel endpoint conditions, so this decomposition is measurable. Since d is 1-Lipschitz and $0 \le d \le 1$, a close pair through z satisfies $|d(\zeta) - d(z)| \le |x-y| \le \delta d(z)$ at every point ζ of the joining segment. Hence

$$(1-\delta)d(z) \le d(\zeta) \le (1+\delta)d(z).$$

In particular, the segment lies in B_2, and the estimates (4.2)–(4.3) apply there with comparable powers of $d(z)$. All comparison constants below are uniform for $0 < \delta < 1/4$, since $3/4 \le d(\zeta)/d(z) \le 5/4$. Write

$$\mathcal{R}(x,y) = |\Phi - \eta(x) + w(x)D(x,y)| + |\eta(x) - \eta(y)|.$$

For $x \neq 0$, $\nabla X(x) = \eta(x)\,\mathrm{Id} - w(x)\omega_x\otimes\omega_x$. At zero, $\nabla X(0) = \mathrm{Id}$, so the same formula holds with our convention for ω_0. For a nondiagonal pair, put $\nu = (x-y)/|x-y|$. The segment formula

$$\Phi(x,y) = \int_0^1 \nu\cdot\nabla X(x - t(x-y))\nu\,dt$$

and $\nu \cdot \nabla X(x)\nu = \eta(x) - w(x)D(x,y)$ give, on a close pair through z,

$$|\Phi - \eta(x) + w(x)D(x,y)| \le C|x-y|d(z)^{L_0-2} \le C\delta d(z)^{L_0-2+j_0},$$
$$|\eta(x) - \eta(y)| \le C|x-y|d(z)^{L_0-1} \le C\delta d(z)^{L_0-1+j_0}.$$

Since $j_0 \ge 2$, both powers are at least L_0. The endpoint comparison and $d^{L_0} \le \eta$ yield the pointwise bound

$$\mathcal{R}(x,y) \le C\delta(\eta(x)+\eta(y)) \quad \text{on the close set.}$$

This bound holds for any witnessing endpoint and therefore gives the same measurable majorant without selecting such an endpoint. The Tonelli identities then give $\iint_{\text{close}} \mathcal{R}\, d\pi \le C\delta E$.

For a far pair,

$$\mathcal{R}(x,y) \le C \sum_{\substack{z\in\{x,y\}\\ d(z)>0}} \left(d(z)^{L_0-1} + \frac{d(z)^{L_0}}{\ell(z)} \right).$$

Indeed, $|\Phi| \le C(\eta(x)+\eta(y))/|x-y|$, while the remaining terms are bounded by $C(d(x)^{L_0-1} + d(y)^{L_0-1})$. A far pair has $|x-y| > \ell(z)$ for every endpoint with $d(z) > 0$. The x-endpoint contribution is therefore controlled by the truncated-potential estimate of Lemma 3.1, which gives the bound

$$C\int_{B_2} f(x)\left[\delta^{-\lambda} d(x)^{L_0-1-j_0\lambda} + \delta^{-\lambda-1} d(x)^{L_0-j_0(\lambda+1)}\right] dx.$$

The two powers are, respectively,

$$L_0 - 1 - j_0\lambda = j_0\sigma + 2N + 1 > 3N + 4, \qquad L_0 - j_0(\lambda+1) = j_0(\sigma-1) + 2N + 2 > 2N + 2.$$

They are nonnegative, and $0 \le d \le 1$, so the universal source mass bound makes this integral at most C_δ. The y-endpoint is handled in the same way with g.

Finally, $\iint |\eta(x) - \eta(y)|\, d\pi \le E_f + E_g < \infty$. Subtracting the two finite Tonelli identities therefore gives the exact formula

$$E_f - E_g = \iint (\eta(x) - \eta(y))\, d\pi.$$

Consequently the left side of (4.7) is at most $\iint \mathcal{R}\, d\pi \le C\delta E + C_\delta$. The coefficient C is independent of δ and the pair, and C_δ is independent of the pair by the universal source and truncated-potential bounds. Choose $0 < \delta < 1/4$ so that $C\delta \le \varepsilon$, and take $C_\varepsilon = C_\delta$. □

5. Interval comparison and source-layer pressure

The radial loss w in the virial identity will be estimated by slicing the source level sets along lines. The interval comparison below retains the coefficient λ of the kernel homogeneity; no special value of this coefficient is required.

Lemma 5.1 (Endpoint comparison on two intervals)**.** *Let $r, \lambda, c > 0$, and define*

$$K_r(s,s') = c\big(r^2 + (s-s')^2\big)^{-\lambda/2}, \qquad D_r(s,s') = \frac{(s-s')^2}{r^2 + (s-s')^2}.$$

Let I, J be nonempty open intervals and $c_I, c_J \ge 0$. Assume that I is bounded when $c_I > 0$, and that J is bounded when $c_J > 0$. For a bounded interval $I = (i_-, i_+)$, put

$$K_{r,I}(s') = \tfrac{1}{2}\big(K_r(i_-,s') + K_r(i_+,s')\big),$$

and define $K_{r,J}(s)$ analogously. Then, as an inequality of extended nonnegative integrals,

$$\int_{I\times J} (c_I + c_J)K_r \le \int_{I\times J} (c_I K_{r,I} + c_J K_{r,J}) + \int_{I\times J} [\lambda \min\{c_I, c_J\} D_r + |c_I - c_J|] K_r. \tag{5.1}$$

A zero endpoint coefficient means a zero term, without evaluating the corresponding endpoints.

Proof. If at most one coefficient is positive, the mismatch term already equals the left integrand. Suppose both coefficients are positive, so both intervals are bounded. All integrals are then finite because $r > 0$. Let m_I, m_J be their midpoints, $d_0 = m_I - m_J$, and $h = s - s'$. The kernel derivatives are

$$\partial_s K_r = -\lambda \frac{h}{r^2+h^2} K_r, \qquad \partial_{s'} K_r = \lambda \frac{h}{r^2+h^2} K_r.$$

Integration by parts in s gives

$$\int_{I\times J} (K_r - K_{r,I}) = -\int_{I\times J} (s - m_I)\partial_s K_r,$$

and the corresponding formula holds in s'. Adding them yields

$$\int_{I\times J} (2K_r - K_{r,I} - K_{r,J}) = \lambda \int_{I\times J} \frac{h^2 - d_0 h}{r^2+h^2} K_r. \tag{5.2}$$

The pushforward of Lebesgue measure on $I \times J$ under $(s, s') \mapsto h$ has density $\rho(h) = |I \cap (J + h)|$: indeed, for fixed h, the permissible values of s are precisely $I \cap (J + h)$. The centers of these two intervals are separated by $|h - d_0|$, so the elementary overlap formula gives $\rho(h) = F(|h - d_0|)$, where

$$F(t) = \max\left\{0, \min\left\{|I|, |J|, \frac{|I| + |J|}{2} - t\right\}\right\}.$$

This is nonincreasing in t. To make the sign explicit, put $k_r(h) = c(r^2 + h^2)^{-\lambda/2}$. Pairing the positive and negative half-lines gives

$$\int_{I\times J} \frac{h}{r^2+h^2} K_r = \int_0^\infty \frac{h\, k_r(h)}{r^2+h^2} \left[F(|h - d_0|) - F(|h + d_0|)\right] dh.$$

If $d_0 \geq 0$, then $|h - d_0| \leq |h + d_0|$ for $h > 0$, so the bracket is nonnegative; if $d_0 \leq 0$, the bracket is nonpositive. Thus

$$d_0 \int_{I\times J} \frac{h}{r^2+h^2} K_r \geq 0.$$

Therefore (5.2) is at most $\lambda \int_{I\times J} D_r K_r$. Put $c_* = \min\{c_I, c_J\}$, and split the weighted defects as

$$\begin{aligned} c_I(K_r - K_{r,I}) + c_J(K_r - K_{r,J}) &= c_*(2K_r - K_{r,I} - K_{r,J}) \\ &\quad + (c_I - c_*)(K_r - K_{r,I}) + (c_J - c_*)(K_r - K_{r,J}). \end{aligned}$$

Each individual defect has integral at most $\int_{I\times J} K_r$, since its endpoint average is nonnegative. The excess coefficients are nonnegative and sum to $|c_I - c_J|$. Integrating the split and then moving the finite endpoint terms to the right proves (5.1). □

Proposition 5.2 (Source-layer pressure)**.** *For every $\varepsilon > 0$ there is a universal constant C_ε such that*

$$aW_f + bW_g \leq \lambda \iint w(x) D(x, y)\, d\pi + \varepsilon E + C_\varepsilon. \tag{5.3}$$

Proof. Let $dS(e)$ be surface measure on $\mathbb{S}^{N-1}$, and put

$$d\nu(e) = \frac{dS(e)}{|\mathbb{S}^{N-1}|}.$$

We use surface integration, polar coordinates and Tonelli's theorem to estimate the directional layers below. The auxiliary parameters will be selected in the order κ, then δ, then P: throughout the proof $0 < \kappa < 1$, $0 < \delta < 1/4$, and $P \geq 1$.

Angular localization. Fix once and for all a smooth function $0 \leq \psi \leq 1$ on $[0, \infty)$, equal to one near zero and supported in $[0, 1]$, independently of the pair and of κ, δ, P. For any unit vector θ, let

$$Z(t) = \int_{\mathbb{S}^{N-1}} \psi(|e - \theta|/t)\, d\nu(e), \qquad 0 < t \leq 1.$$

Rotational invariance makes this independent of θ. If φ is the polar angle about θ, the chordal radius $\rho = 2\sin(\varphi/2)$ satisfies $\sin\varphi = \rho(1-\rho^2/4)^{1/2}$ and $d\varphi = (1-\rho^2/4)^{-1/2}\,d\rho$. Substituting $\rho = ts$ in the surface integral gives, also when $N = 2$ with $|\mathbb{S}^0| = 2$,

$$Z(t) = \frac{|\mathbb{S}^{N-2}|}{|\mathbb{S}^{N-1}|} t^{N-1} \int_0^1 \psi(s) s^{N-2} \left(1 - \frac{t^2 s^2}{4}\right)^{(N-3)/2} ds. \tag{5.4}$$

For $0 < t \le 1$ and $0 \le s \le 1$, the base $1 - t^2 s^2/4$ lies in $[3/4, 1]$. The integral factor in (5.4) is therefore bounded above and below by positive constants, and its t-derivative is $O(t)$. Differentiation under the integral is justified by these uniform bounds. Consequently $c_Z t^{N-1} \le Z(t) \le C_Z t^{N-1}$ and $|Z'(t)| \le C_Z t^{N-2}$. Define

$$h_e(x) = \frac{w(x)}{Z(\kappa d(x))} \psi\left(\frac{|e - \omega_x|}{\kappa d(x)}\right) \quad (1 < |x| < 2), \tag{5.5}$$

and set it to zero elsewhere. Then

$$\begin{gathered} \int_{\mathbb{S}^{N-1}} h_e(x)\, d\nu(e) = w(x), \\ 0 \le h_e(x) \le C_\kappa d(x)^{L_0 - N}, \qquad |\nabla h_e(x)| \le C_\kappa d(x)^{L_0 - N - 1}. \end{gathered} \tag{5.6}$$

The profile is constant near the center of the cap, which removes the apparent derivative singularity of the chordal norm there. More explicitly, put $\rho_e(x) = |e - \omega_x|/(\kappa d(x))$. On the support of $\psi'(\rho_e)$, the bound $|e - \omega_x| \le \kappa d(x)$ and the bounded derivative of ω_x on the annulus give $|\nabla \rho_e| \le C/(\kappa d(x))$. Also

$$\frac{1}{Z(\kappa d)} \le C \kappa^{-(N-1)} d^{-(N-1)}, \qquad \frac{\kappa |Z'(\kappa d)|}{Z(\kappa d)} \le \frac{C}{d}.$$

Differentiating (5.5), using $|\nabla w| \le C d^{L_0 - 2}$ and $|\nabla d| = 1$ on the annulus, proves the displayed gradient estimate, uniformly in e. More precisely, the two bounds can be taken with factors $C\kappa^{-(N-1)}$ and $C\kappa^{-N}$, respectively. At the inner junction, the formulas in Section 4 give $w = O((|x|-1)^2)$ and $|\nabla w| = O(|x|-1)$. The remaining factors and their x-derivatives are bounded for fixed κ, uniformly in e. Thus $h_e = O_\kappa((|x|-1)^2)$ and $|\nabla h_e| = O_\kappa(|x|-1)$ there. At the outer junction the displayed powers tend to zero because $L_0 - N > 1$. It follows that h_e extends by zero to a C^1 function on $\mathbb{R}^N$. The same uniform bounds show that $(e, x) \mapsto h_e(x)$ is jointly continuous.

Apply the measurable close/far decomposition and segment comparison from Lemma 4.2, with the present δ, to h_e. The close-pair power is $L_0 - N - 1 + j_0 = L_0 + 2$, so its contribution is at most $C_\kappa \delta E$. On far pairs, use $h_e(x) + h_e(y)$ and the truncated-potential bound (3.4). The two endpoint contributions are at most

$$C_\kappa \delta^{-\lambda} \int_{B_2} d(x)^{L_0 - N - j_0\lambda} (f(x) + g(x))\, dx \le C_{\kappa,\delta},$$

because $L_0 - N - j_0\lambda = N + 2 + j_0\sigma > 0$ and the source masses are universal. Hence, uniformly in e,

$$\iint |h_e(x) - h_e(y)|\, d\pi \le C_\kappa \delta E + C_{\kappa,\delta}. \tag{5.7}$$

On the support of $h_e(x)$, $|e - \omega_x| \le \kappa d(x)$. For unit vectors, $e \mapsto D_e(x, y)$ is Lipschitz with constant at most two. The exact angular normalization therefore gives the pointwise estimate, off the diagonal,

$$\left| \int_{\mathbb{S}^{N-1}} h_e(x) D_e(x, y)\, d\nu(e) - w(x) D(x, y) \right| \le 2\kappa d(x) w(x) \le C\kappa \eta(x).$$

Integrating against π and using its x-marginal gives

$$\int_{\mathbb{S}^{N-1}} \left[\iint h_e(x) D_e(x, y)\, d\pi \right] d\nu(e) \le \iint w(x) D(x, y)\, d\pi + C\kappa E. \tag{5.8}$$

Retaining source layers. For $t > 0$, intersect the open set $\{f > t\}$ with lines parallel to e, and let I denote one of its interval components. The definition $f(0) = 0$ excludes the origin from every such set; its openness follows from continuity away from zero. Assign to I the coefficient

$$\chi_I = \begin{cases} \inf_I h_e, & I \text{ is bounded and } |I| \le 2\delta \inf_I d, \\ 0, & \text{otherwise.} \end{cases}$$

Write $\chi_f(e,x,t) = \chi_I$ when $x \in I$, and zero when $f(x) \le t$. Define χ_g in the same way. In particular, $0 \le \chi_f(e,x,t) \le h_e(x)$.

These coefficients and the endpoints used below are jointly measurable. For $0 < t < f(x)$, define the exit distances

$$a_f^{\pm}(e,x,t) = \sup\{r \ge 0 : x \pm [0,r]e \subset \{f > t\}\}.$$

For every fixed $r \ge 0$, the event $a_f^+ > r$ is precisely containment of $x + [0,r]e$ in the layer. This is open in (e,x,t): the compact segment avoids zero, and continuity gives a positive margin above t on it. The backward event is identical. Extend both exit distances by zero outside the layer; they are thus measurable, with infinity allowed. The component through x is

$$I_f(e,x,t) = \{x + se : -a_f^- < s < a_f^+\}, \qquad |I_f| = a_f^- + a_f^+.$$

This formula gives the empty set when $f(x) \le t$. For the infimum maps, use $\inf_\varnothing d = \inf_\varnothing h_e = +\infty$; the coefficient is still defined to be zero outside the layer. The infima of d and h_e on this component can be taken over rational offsets s in the indicated open interval. Continuity of d and joint continuity of h_e therefore make these infima measurable. On bounded components the endpoints $x_- = x - a_f^- e$, $x_+ = x + a_f^+ e$ are measurable as well. Consequently the retention test, coefficient, and every endpoint term with positive coefficient are measurable. The same reasoning applies to g. All nonnegative level, spatial, and surface integrals below may therefore be taken by Tonelli.

For later use, put

$$H_f(e) = \int h_e f u, \qquad H_g(e) = \int h_e g v.$$

These are finite for each direction, since h_e is bounded and supported in the compact annulus and the local energy densities are integrable. Angular normalization and Tonelli give $\int H_f \, d\nu = W_f$ and $\int H_g \, d\nu = W_g$.

The discarded parts of the layers contribute the nonnegative losses

$$L_f(e) = \int u(x) \int_0^{f(x)} (h_e(x) - \chi_f(e,x,t)) \, dt \, dx,$$

and $L_g(e)$ with v, g. Thus $0 \le L_f(e) \le H_f(e)$ and $0 \le L_g(e) \le H_g(e)$. We claim

$$\int_{\mathbb{S}^{N-1}} (L_f(e) + L_g(e)) \, d\nu(e) \le \left(C_\kappa \delta + \frac{C_{\kappa,\delta}}{P} \right) E + CP. \tag{5.9}$$

Here the constant in CP is independent of κ, δ, P, and $C_{\kappa,\delta}$ is independent of P; all constants are independent of the pair.

It suffices to consider x with $1 < |x| < 2$; elsewhere $h_e(x) = 0$. Set $d = d(x)$ and $\ell = \delta d^{j_0}$. If its component has length at most ℓ, then every $y \in I$ satisfies $(1-\delta)d \le d(y) \le (1+\delta)d$, by the segment comparison of Section 4. In particular

$$|I| \le \ell \le \delta d \le 2\delta(1-\delta)d \le 2\delta \inf_I d.$$

The component therefore passes the length test. The directional gradient bound, applied along I, gives

$$h_e(x) - \chi_f(e,x,t) \le C_\kappa \delta d^{L_0 - N - 1 + j_0} \le C_\kappa \delta d^{L_0}.$$

Integrating this bound over directions and the high levels $Pd^{-k_0} \le t < f(x)$, and then against $u(x)\,dx$, contributes at most $C_\kappa \delta E_f$, since $d^{L_0} \le \eta(x)$.

The contribution of long components admits a direct cone estimate. If $|I_f(e,x,t)| > \ell$, one of the two exit distances from x exceeds $\ell/2$, including when the component is unbounded. On the corresponding radial segment the source exceeds t. Consequently, for every direction e,

$$\mathbf{1}_{\{|I_f(e,x,t)|>\ell\}} \le \frac{N}{t(\ell/2)^N} \int_0^{\ell/2} \big[f(x+re) + f(x-re)\big] r^{N-1}\, dr. \tag{5.10}$$

Multiply by $h_e(x)$, integrate in e, and use polar coordinates centered at x. With $\omega_z = z/|z|$ as before, this gives

$$\begin{aligned} &\int_{\mathbb{S}^{N-1}} h_e(x) \mathbf{1}_{\{|I_f(e,x,t)|>\ell\}}\, d\nu(e) \\ &\quad \le \frac{2^N N}{|\mathbb{S}^{N-1}|\, t\ell^N} \int_{B_{\ell/2}(x)} f(y) \big[h_{\omega_{y-x}}(x) + h_{-\omega_{y-x}}(x)\big]\, dy \\ &\quad \le \frac{2^{N+1} N \sup_e h_e(x)}{|\mathbb{S}^{N-1}|\, t\ell^N} \int_{B_{\ell/2}(x)} f(y)\, dy \le \frac{C}{t\ell^N} \sup_e h_e(x). \end{aligned} \tag{5.11}$$

The last step uses $B_{\ell/2}(x) \subset B_4$ and the universal source mass bound. The value of ω_{y-x} at $y = x$ is immaterial. The backward segment produces $h_{-\omega_{y-x}}(x)$ after the change $e \mapsto -e$; thus both exit directions are included in the same nonnegative polar integral.

For levels $t \ge Pd^{-k_0}$, the surface integral of the long-component loss is bounded by the left side of (5.11). Substituting $\ell = \delta d^{j_0}$ and $\sup_e h_e(x) \le C_\kappa d^{L_0-N}$ gives

$$\begin{aligned} &\int_{\mathbb{S}^{N-1}} (h_e(x) - \chi_f(e,x,t)) \mathbf{1}_{\{|I_f(e,x,t)|>\ell\}}\, d\nu(e) \\ &\qquad \le \frac{C_{\kappa,\delta}}{P} d^{L_0-N+k_0-j_0N} = \frac{C_{\kappa,\delta}}{P} d^{L_0}. \end{aligned}$$

Here $k_0 = N(j_0+1)$. Integration against $u(x)\,dt\,dx$ gives $C_{\kappa,\delta} P^{-1} E_f$. For the remaining levels, the exact angular normalization gives

$$\begin{aligned} &\int_0^{\min\{f(x), Pd^{-k_0}\}} \int_{\mathbb{S}^{N-1}} (h_e(x) - \chi_f(e,x,t))\, d\nu(e)\, dt \\ &\qquad \le Pd^{-k_0} w(x) \le CPd^{L_0-1-k_0} \le CP. \end{aligned}$$

Here $L_0 - 1 - k_0 = N + 1 > 0$; the universal bound for $\int_{B_4} u$ therefore gives a contribution at most CP. The same argument for g, v proves (5.9).

Weighted endpoints and interval integration. If $\chi_I > 0$, the closure of I lies in the annulus $1 < |x| < 2$: I is bounded, $h_e \ge \chi_I > 0$ on I, and continuity preserves this inequality at its endpoints. Since $h_e = 0$ outside this open annulus, its positive lower bound places the entire closed interval inside the annulus. Continuity of f gives $f(x_\pm) \ge t$, while $f(x_\pm) > t$ would extend the open component past its endpoint. Thus $f(x_\pm) = t$. Let

$$u_I = \tfrac{1}{2}(u(x_-) + u(x_+)), \qquad \theta_f(x,t) = \left(\frac{t}{|x|^B}\right)^{1/q}.$$

Define this ideal endpoint value for $x \ne 0$; all level integrals below disregard the singleton at zero. Use the same convention for θ_g. On the layer, $0 < \theta_f(x,t) < u(x)$, and the endpoint values are $u(x_\pm) = t^{1/q}|x_\pm|^{-B/q}$. Moreover, $|x_\pm - x| \le 2\delta d(x)$. For $H(x) = |x|^{-B/q}$, $\nabla \log H(x) = -(B/q)x/|x|^2$, whose norm is at most $|B|/q$ on the annulus. The segments from x to its endpoints lie in the closure of the retained interval and hence in the annulus. Integrating this logarithmic gradient therefore gives

$$\left|\frac{H(x_\pm)}{H(x)} - 1\right| \le e^{2|B|\delta d(x)/q} - 1 \le C\delta d(x).$$

Multiplication by $t^{1/q} H(x)$ and averaging the endpoints yield

$$|u_I - \theta_f(x,t)| \le C\delta d(x)\theta_f(x,t) \le C\delta d(x) u(x). \tag{5.12}$$

For a retained g-component J at level t', define the endpoint average v_J and $\theta_g(y,t') = (t'/|y|^A)^{1/p}$. They obey the same estimate. Whenever a retained coefficient is zero, its endpoint term is defined as zero before any endpoint is evaluated.

Fix e, and write $x = z+se$, $y = z'+s'e$, with $z, z' \in e^\perp$. For $z \neq z'$, apply Lemma 5.1 to every pair of components of the two source layers, using $r = |z - z'|$ and the coefficients χ_I, χ_J. The line kernel is exactly $K(x-y)$, and $D_r = D_e(x,y)$. For this fixed e, the linear coordinates have Jacobian one. Sum over the countably many components and integrate over z, z' and the levels t, t'. No measurable enumeration of the components is needed: at each point the coefficients and endpoints are the measurable functions of the exit maps constructed above. Endpoint terms with zero coefficient are set to zero before evaluation. Thus each term is a nonnegative measurable integrand, and countable additivity followed by Tonelli applies. Subsequent integration in e is performed in these original pointwise variables; it does not require a continuously chosen transverse coordinate frame. The transverse diagonal $z = z'$ is null because $N - 1 \geq 1$. For each fixed x, or each of its retained endpoints, its omission removes only the affine line $x + \mathbb{R}e$ from the partner integral. This line is Lebesgue null in $\mathbb{R}^N$, so both the ordinary and endpoint potentials are recovered exactly. For example, at a retained f-component,

$$\int_{\mathbb{R}^N} \int_0^{g(y)} \frac{K(x_- - y) + K(x_+ - y)}{2}\, dt'\, dy = \frac{u(x_-) + u(x_+)}{2} = u_I.$$

The exact potentials similarly turn the other endpoint kernels into v_J; the retained left terms recover $u(x)$ and $v(y)$.

Before making any subtraction, we verify that every resulting integral is finite. The two retained left terms are at most $H_f(e)$ and $H_g(e)$. By (5.12) and $\theta_f \leq u$, the endpoint terms are at most $CH_f(e)$ and $CH_g(e)$. The pressure term is bounded by $\lambda H_f(e)$, and the mismatch term by $H_f(e) + H_g(e)$. The convention for zero coefficients avoids any need to subtract infinite unweighted line integrals.

We may consequently define the signed, absolutely integrable quantity

$$\begin{aligned} S_e = &\int \int_0^{f(x)} \chi_f(e,x,t)(u(x) - u_I)\, dt\, dx \\ &+ \int \int_0^{g(y)} \chi_g(e,y,t')(v(y) - v_J)\, dt'\, dy. \end{aligned}$$

To estimate the coefficient mismatch in the integrated interval inequality, use

$$|\chi_f - \chi_g| \leq |h_e(x) - h_e(y)| + (h_e(x) - \chi_f) + (h_e(y) - \chi_g).$$

At fixed (x,y), the level integrals satisfy

$$\begin{aligned} &\int_0^{f(x)} \int_0^{g(y)} \min\{\chi_f(e,x,t), \chi_g(e,y,t')\}\, dt'\, dt \leq h_e(x) f(x) g(y), \\ &\int_0^{f(x)} \int_0^{g(y)} (h_e(x) - \chi_f(e,x,t))\, dt'\, dt = g(y) \int_0^{f(x)} (h_e(x) - \chi_f(e,x,t))\, dt. \end{aligned}$$

Multiplying the second identity by $K(x-y)$ and integrating first in y gives exactly $L_f(e)$; reversing the roles gives $L_g(e)$. The mismatch term independent of the levels integrates to $\iint |h_e(x) - h_e(y)|\, d\pi$. We obtain

$$S_e \leq \lambda \iint h_e(x) D_e(x,y)\, d\pi + \iint |h_e(x) - h_e(y)|\, d\pi + L_f(e) + L_g(e). \tag{5.13}$$

For $x \neq 0$, direct integration gives $\int_0^{f(x)} \theta_f(x,t)\, dt = qf(x)u(x)/(q+1)$. The analogous identity for g has p in place of q. Hence the ideal weighted layer differences have the exact integrals

$$\begin{aligned} &\int_0^{f(x)} (u(x) - \theta_f(x,t))\, dt = af(x)u(x), \\ &\int_0^{g(y)} (v(y) - \theta_g(y,t'))\, dt' = bg(y)v(y). \end{aligned}$$

For an f-layer, including the zero-coefficient case,

$$\begin{aligned} h_e(u-\theta_f) - \chi_f(u-u_I) &= (h_e-\chi_f)(u-\theta_f) + \chi_f(u_I-\theta_f) \\ &\le (h_e-\chi_f)u + C\delta d\, h_e u. \end{aligned}$$

Only $u-\theta_f$ is assumed nonnegative; the retained difference $u-u_I$ may have either sign. After integrating this inequality and its g-counterpart, we find

$$aH_f(e) + bH_g(e) \le S_e + L_f(e) + L_g(e) + C\delta \int dh_e f u + C\delta \int dh_e g v. \tag{5.14}$$

The signed quantity S_e is also absolutely integrable in the direction variable: the preceding bounds for the retained and endpoint terms, together with the previously established averages of H_f, H_g, give

$$\int_{\mathbb{S}^{N-1}} |S_e|\, d\nu(e) \le C(W_f + W_g) < \infty.$$

Thus we may integrate (5.13) and (5.14) against $d\nu(e)$. Combining them incurs two copies of $L_f + L_g$. Equations (5.6)–(5.8), (5.9), and $dw \le C\eta$ give

$$aW_f + bW_g \le \lambda \iint w(x)D(x,y)\, d\pi + \left(C\kappa + C\delta + C_\kappa\delta + \frac{C_{\kappa,\delta}}{P}\right) E + C_{\kappa,\delta,P}.$$

First choose κ so that $C\kappa \le \varepsilon/3$. With κ fixed, choose $0 < \delta < 1/4$ so that $(C + C_\kappa)\delta \le \varepsilon/3$; finally choose $P \ge 1$ so large that $C_{\kappa,\delta}/P \le \varepsilon/3$. All choices depend only on ε and the fixed system parameters. The coefficient of E is therefore at most ε, and the remaining constant is universal, proving (5.3). □

6. The energy bound and the scaling contradiction

Theorem 6.1. *In the potential setting of Section 3, if*

$$\frac{N+A}{p+1} + \frac{N+B}{q+1} > N - \sigma, \tag{6.1}$$

there is no finite positive continuous pair satisfying (3.1).

Proof. Suppose for contradiction that such a pair exists. Use the notation of Sections 4 and 5, and set

$$T = \iint w(x)D(x,y)\, d\pi, \qquad Q = \iint (\Phi - \eta(x) + w(x)D(x,y))\, d\pi.$$

The first integral satisfies $0 \le T \le W_f < \infty$, since $0 \le D \le 1$ and the x-marginal of π is fu. Lemma 4.2 proves absolute convergence of Q. The finite Tonelli identity $\iint \eta(x)\, d\pi = E_f$ consequently gives

$$\iint \Phi\, d\pi = E_f - T + Q.$$

With

$$c_1 = a(N+B) - \lambda, \qquad c_2 = b(N+A), \qquad \gamma = c_1 + c_2 = a(N+B) + b(N+A) - \lambda > 0,$$

where positivity follows from (6.1), the virial identity becomes

$$c_1E_f + c_2E_g = aW_f + bW_g - \lambda T + \lambda Q.$$

For any $\varepsilon > 0$, take a common positive universal constant C_ε large enough for Proposition 5.2 and Lemma 4.2. They give

$$\begin{aligned} aW_f + bW_g - \lambda T &\le \varepsilon E + C_\varepsilon, \\ |Q| + |E_f - E_g| &\le \varepsilon E + C_\varepsilon. \end{aligned}$$

The second bound uses $|Q| \le \iint |\Phi - \eta(x) + w(x)D|\, d\pi$. To handle either sign of c_1, decompose

$$c_1E_f + c_2E_g = \frac{\gamma}{2}E + \frac{c_1 - c_2}{2}(E_f - E_g).$$

Putting $M = 1 + \lambda + |c_1 - c_2|/2$, we therefore obtain

$$\begin{aligned}\frac{\gamma}{2}E &\le \varepsilon E + C_\varepsilon + \lambda|Q| + \frac{|c_1 - c_2|}{2}|E_f - E_g| \\ &\le M(\varepsilon E + C_\varepsilon).\end{aligned}$$

Choose $\varepsilon_* = \gamma/(4M) > 0$. Since E is finite, subtracting $\gamma E/4$ gives

$$E \le \frac{4M}{\gamma} C_{\varepsilon_*} =: C.$$

Here M, γ, ε_* depend only on the fixed system parameters, so C is universal. As $\eta = 1$ on B_1 and all energy densities are nonnegative,

$$\int_{B_1} \left(|x|^B u^{q+1} + |x|^A v^{p+1}\right) dx \le E \le C. \tag{6.2}$$

Section 3 already shows that every dilated pair u_R, v_R in (3.6) satisfies the same potential system with the same fixed parameters. Thus (6.2) applies for every $R > 0$ with the same constant C. The scaling relations (2.2) give

$$\alpha(q+1) - B - N = \beta(p+1) - A - N = \alpha + \beta + \sigma - N =: \tau.$$

Changing variables $z = Rx$ therefore yields

$$\begin{aligned}&\int_{B_1} \left(|x|^B u_R^{q+1} + |x|^A v_R^{p+1}\right) dx \\ &\qquad = R^\tau \int_{B_R} \left(|z|^B u^{q+1} + |z|^A v^{p+1}\right) dz.\end{aligned}$$

The identity (2.8) is purely algebraic and applies to the present σ. Hence

$$\gamma = (1 - a - b)\tau, \qquad 1 - a - b = \frac{pq - 1}{(p+1)(q+1)} > 0,$$

so $\tau > 0$. Put

$$I_1 = \int_{B_1} \left(|x|^B u^{q+1} + |x|^A v^{p+1}\right) dx.$$

This number is finite by local boundedness and $A, B > -N$. It is strictly positive because the pair and weights are positive on $B_1 \setminus \{0\}$. For $R \ge 1$, nonnegativity of the energy density and the bound for the dilated pair give

$$C \ge R^\tau \int_{B_R} \left(|z|^B u(z)^{q+1} + |z|^A v(z)^{p+1}\right) dz \ge R^\tau I_1 \longrightarrow \infty.$$

This contradiction proves the theorem. □

Proof of Theorem 1.1. Suppose the pair in Theorem 1.1 is nonzero. Proposition 2.3 gives a finite positive continuous pair of exact potentials and forces $A, B > -2m$. Take $\sigma = 2m$ and $c = c_{N,2m} > 0$. Since $m \ge 1$ and $N > 2m$, we have $1 < \sigma < N$; the exponent assumptions are unchanged, and (1.2) is exactly (6.1). Thus every hypothesis of Theorem 6.1 holds, giving a contradiction. Therefore $u \equiv v \equiv 0$. □

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